Weather Data Analysis — Python Code

Requirements: pandas, matplotlib, and a weather.csv file with Date, Temperature, and Humidity columns.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load weather data
df = pd.read_csv("weather.csv")

# 1. Average, maximum and minimum temperature
print("Average Temperature:", df["Temperature"].mean())
print("Maximum Temperature:", df["Temperature"].max())
print("Minimum Temperature:", df["Temperature"].min())

# 2. Temperature changes (Line Chart)
plt.plot(df["Date"], df["Temperature"], marker="o")
plt.title("Temperature Changes")
plt.xlabel("Date")
plt.ylabel("Temperature")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 3. Temperature distribution (Histogram)
plt.hist(df["Temperature"], edgecolor="black")
plt.title("Temperature Distribution")
plt.xlabel("Temperature")
plt.ylabel("Frequency")
plt.show()

# 4. Day with highest temperature
hottest = df.loc[df["Temperature"].idxmax()]
print("\\nHottest Day:")
print(hottest)

# 5. Humidity vs Temperature
plt.scatter(df["Humidity"], df["Temperature"])
plt.title("Humidity vs Temperature")
plt.xlabel("Humidity")
plt.ylabel("Temperature")
plt.show()

# 6. Temperature vs Date (Scatter Plot)
plt.scatter(df["Date"], df["Temperature"])
plt.title("Temperature vs Date")
plt.xlabel("Date")
plt.ylabel("Temperature")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```